

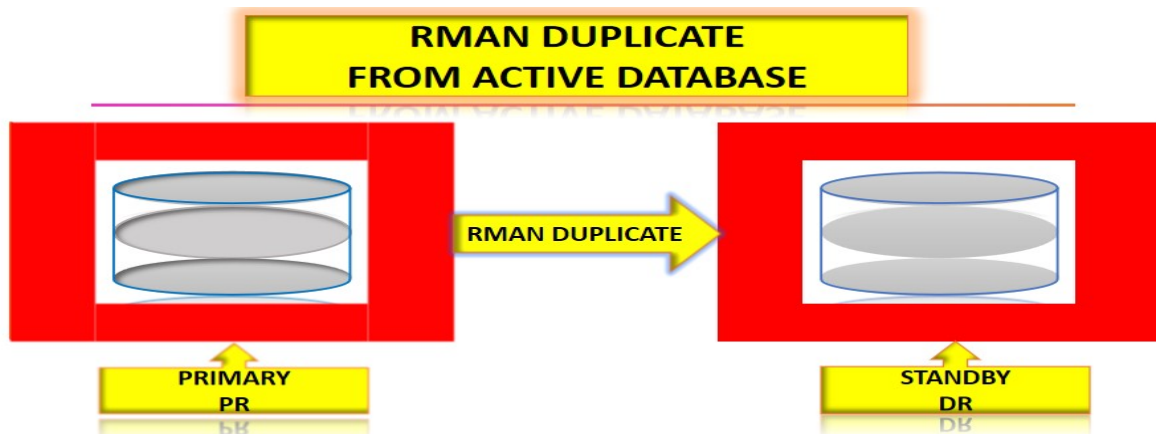
# Active Dataguard in CDB on Non-ASM [19.0.0]

## Creating Physical Standby Active DataGuard Configuration in 19.0.0 Container Database

**AIM:** Without shutting down primary, we need to create physical standby database using RMAN DUPLICATE FROM ACTIVE DATABASE command on 19.0.0 Database in CDB/PDB environment (No need to take backup of primary database)

### Contents

---



### 0. Introduction

1. Environment
2. Enable Forced Logging on Primary
3. Configure a Standby Redo Log on Primary
4. Verify Archive Mode Enabled on Primary
5. Set Primary Database Initialization Parameters
6. Transfer PFILE AND PASSWORD FILE TO STANDBY SIDE
7. Prepare an Initialization Parameter File for the Standby Database on Standby
8. Create required directories on Standby
9. Add below entry in ORATAB on Standby
10. Create spfile from pfile on Standby
11. Configure TNS Entries on Primary
12. Configure TNS Entries on Standby
13. Verify connection 'AS SYSDBA' from Primary
14. Create Standby Database using duplicate command
15. Enable MRP on Standby
16. Open PDB in READ-ONLY Mode
17. Verify the log shipping is performing properly
18. Let's check that primary and standby DBs are in sync
19. Switchover status from Primary database

### 0. Introduction

PLEASE NOTE in 19c Data Guard is set up at the Container level and not the individual Pluggable database level as the redo log files only belong to the Container database and the individual pluggable databases do not have their own online redo log files.

**Definition of Active Dataguard:** Oracle Active Data Guard enables read-only access to a physical standby database for queries, sorting, reporting, web-based access, etc., while continuously applying changes received from the production/primary database.

## 1. Environment

Source:

| PRIMARY (node1)            |                                                           |
|----------------------------|-----------------------------------------------------------|
| Platform                   | Linuxx86_64   Red Hat Enterprise Linux Server release 7.8 |
| DB Version & Database type | Oracle 19.0.0 & CDB-PDB                                   |
| db_name                    | ORCLPR                                                    |
| CDB-PDB Name               | PRODPDB                                                   |
| DB_UNIQUE_NAME             | ORCLPR                                                    |
| service_name               | orclpr                                                    |

Target:

| STANDBY (node2)            |                                                           |
|----------------------------|-----------------------------------------------------------|
| Platform                   | Linuxx86_64   Red Hat Enterprise Linux Server release 7.8 |
| DB Version & Database type | Oracle 19.0.0 & CDB-PDB                                   |
| db_name                    | ORCLPR                                                    |
| CBD-PDB Name               | PRODPDB                                                   |
| DB_UNIQUE_NAME             | ORCL_DR                                                   |
| service_name               | ORCL_DR                                                   |

## 2. Enable Forced Logging on Primary:

```
SQL> select name, open_mode, cdb from v$database;
```

| NAME   | OPEN_MODE  | CDB |
|--------|------------|-----|
| ORCLPR | READ WRITE | YES |

```
SQL> show con_name
```

| CON_NAME  |
|-----------|
| CDB\$ROOT |

```
SQL> col name for a30
```

```
SQL> select name, open_mode, con_id, dbid from v$containers;
```

| NAME      | OPEN_MODE  | CON_ID | DBID       |
|-----------|------------|--------|------------|
| CDB\$ROOT | READ WRITE | 1      | 829140703  |
| PDB\$SEED | READ ONLY  | 2      | 3650434618 |
| PRODPDB   | READ WRITE | 3      | 3936623554 |

```
SQL> select force_logging from v$database;
```

| FORCE_LOGGING |
|---------------|
| NO            |

```
SQL> ALTER DATABASE FORCE LOGGING;
```

Database altered.

```
SQL> select force_logging from v$database;
```

## FORCE\_LOGGING

YES <----

SQL>

```
oracle@pr:~  
SQL>  
SQL>  
SQL> startup mount;  
ORACLE instance started.  
  
Total System Global Area 1241510080 bytes  
Fixed Size          9134272 bytes  
Variable Size       318767104 bytes  
Database Buffers    905969664 bytes  
Redo Buffers        7639040 bytes  
Database mounted.  
SQL>  
SQL> select name, open_mode, cdb from v$database;  
  
NAME          OPEN_MODE          CDB  
-----  
ORCLPR        MOUNTED            YES  
  
SQL> show con_name  
  
CON_NAME  
-----  
CDB$ROOT  
SQL> col name for a30  
SQL> select name, open_mode, con_id, dbid from v$containers;  
  
NAME          OPEN_MODE          CON_ID          DBID  
-----  
CDB$ROOT      MOUNTED            1 829140703  
PDB$SEED      MOUNTED            2 3650434618  
PRODPDB       MOUNTED            3 3936623554  
  
SQL> select force_logging from v$database;  
  
FORCE_LOGGING  
-----  
NO  
  
SQL> ALTER DATABASE FORCE LOGGING;  
  
Database altered.  
  
SQL> select force_logging from v$database;  
  
FORCE_LOGGING  
-----  
YES  
  
SQL> █
```

### 3. Configure a Standby Redo Log on Primary:

Note :

- Since we have 3 online redo log file groups, we need to create 4(3+1) Standby redo log file groups
- Standby Redo logs files come into picture only when protection mode is Maximum Availability and Maximum Protection.

```
oracle@pr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
SQL>
SQL>
SQL>
SQL> show con_name

CON_NAME
-----
CDB$ROOT
SQL>
SQL>
SQL> set lines 180
SQL> col MEMBER for a60
SQL> select b.thread#, a.group#, a.member, b.bytes FROM v$logfile a, v$log b WHERE a.group# = b.group#;

  THREAD#    GROUP# MEMBER                                     BYTES
-----
1         3 /u01/app/oracle/oradata/ORCLPR/online/og/o1_mf_3_m2f8w9ks_1 209715200
1         3 /u01/app/oracle/fast_recovery_area/ORCLPR/online/og/o1_mf_3_m2f8wdjt_1 209715200
1         2 /u01/app/oracle/oradata/ORCLPR/online/og/o1_mf_2_m2f8w9gr_1 209715200
1         2 /u01/app/oracle/fast_recovery_area/ORCLPR/online/og/o1_mf_2_m2f8wdgg_1 209715200
  THREAD#    GROUP# MEMBER                                     BYTES
-----
1         1 /u01/app/oracle/oradata/ORCLPR/online/og/o1_mf_1_m2f8w9fj_1 209715200
1         1 /u01/app/oracle/fast_recovery_area/ORCLPR/online/og/o1_mf_1_m2f8wb6s_1 209715200

6 rows selected.

SQL>
```

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 4 ('/u01/app/oracle/oradata/ORCLPR/online/og/o1_mf_4.log')
SIZE 200M;
```

Database altered.

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 5 ('/u01/app/oracle/oradata/ORCLPR/online/og/o1_mf_5.log')
SIZE 200M;
```

Database altered.

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 6 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_6.log')
SIZE 200M;
```

Database altered.

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 7 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_7.log')
SIZE 200M;
```

Database altered.

```
oracle@pr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 4 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_4.log') SIZE 200M;
Database altered.

SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 5 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_5.log') SIZE 200M;
Database altered.

SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 6 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_6.log') SIZE 200M;
Database altered.

SQL> ALTER DATABASE ADD STANDBY LOGFILE GROUP 7 ('/u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_7.log') SIZE 200M;
Database altered.

SQL> select * from v$logfile;

  GROUP# STATUS  TYPE      MEMBER                                                    IS_  CON_ID
-----
      3  ONLINE  ONLINE  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_3_m2f8w9ks_.1 NO      0
      3  ONLINE  ONLINE  /u01/app/oracle/fast_recovery_area/ORCLPR/onlineelog/o1_mf_3_ YES  0
      2  ONLINE  ONLINE  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_2_m2f8w9gr_.1 NO      0
      2  ONLINE  ONLINE  /u01/app/oracle/fast_recovery_area/ORCLPR/onlineelog/o1_mf_2_ YES  0
  GROUP# STATUS  TYPE      MEMBER                                                    IS_  CON_ID
-----
      1  ONLINE  ONLINE  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_1_m2f8w9fj_.1 NO      0
      1  ONLINE  ONLINE  /u01/app/oracle/fast_recovery_area/ORCLPR/onlineelog/o1_mf_1_ YES  0
      4  STANDBY  STANDBY  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_4.log      NO      0
      5  STANDBY  STANDBY  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_5.log      NO      0
      6  STANDBY  STANDBY  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_6.log      NO      0
      7  STANDBY  STANDBY  /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_7.log      NO      0

10 rows selected.

SQL>
```

```
SQL> select a.group#, a.member, b.bytes FROM v$logfile a, v$standby_log b WHERE a.group# = b.group#;
```

| GROUP# | MEMBER                                                | BYTES     |
|--------|-------------------------------------------------------|-----------|
| 4      | /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_4.log | 209715200 |
| 5      | /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_5.log | 209715200 |
| 6      | /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_6.log | 209715200 |
| 7      | /u01/app/oracle/oradata/ORCLPR/onlineelog/o1_mf_7.log | 209715200 |

## 4. Verify Archive Mode Enabled on Primary:

If archive mode is Disabled then, steps to enable archive mode.

First check command: SQL> archive log list;

SQL> archive log list;

|                            |                           |
|----------------------------|---------------------------|
| Database log mode          | No Archive Mode           |
| Automatic archival         | Disabled                  |
| Archive destination        | USE_DB_RECOVERY_FILE_DEST |
| Oldest online log sequence | 7                         |
| Current log sequence       | 9                         |

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_FORMAT='%t\_%s\_%r.dbf' SCOPE=SPFILE;

SQL> ALTER SYSTEM SET log\_archive\_dest\_1='location=/u01/app/oracle/oradata/ORCLPR/archive/' SCOPE=both;

```
SQL*Plus: Release 19.0.0.0.0 - Production on Thu Apr 25 18:01:12 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> ALTER SYSTEM SET log_archive_dest_1='location=/u01/app/oracle/oradata/ORCLPR/archive/' SCOPE=both;

System altered.

SQL> shut immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL>
SQL> startup mount;
ORACLE instance started.

Total System Global Area 1241510080 bytes
Fixed Size 9134272 bytes
Variable Size 335544320 bytes
Database Buffers 889192448 bytes
Redo Buffers 7639040 bytes
Database mounted.
SQL>
SQL> ALTER DATABASE ARCHIVELOG;

Database altered.

SQL>
SQL> archive log list;
Database log mode Archive Mode
Automatic archival Enabled
Archive destination /u01/app/oracle/oradata/ORCLPR/archive/
Oldest online log sequence 7
Next log sequence to archive 9
Current log sequence 9
SQL>
SQL> alter database open;

Database altered.

SQL>
```

## 5. Set Primary Database Initialization Parameters:

SQL> show parameter pfile

| NAME   | TYPE   | VALUE                                                          |
|--------|--------|----------------------------------------------------------------|
| spfile | string | /u01/app/oracle/product/12.1.0.2.0/dbhome_1/dbs/spfileorcl.ora |

SQL>

SQL> create pfile='/u01/app/oracle/product/19.0.0/dbhome\_1/dbs/initORCLPR\_before.ora' from spfile;

File created.

SQL> alter system set db\_unique\_name='ORCLPR' scope=spfile;

System altered.

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_CONFIG='DG\_CONFIG=(ORCLPR,ORCL\_DR)' scope=both;

System altered.

```
SQL> show parameter pfile

NAME                                TYPE                                VALUE
-----                                -                                -
spfile                              string                              /u01/app/oracle/product/12.1.0
                                     .2.0/dbhome_1/dbs/spfileorcl.o
                                     ra

SQL>
SQL> create pfile='/u01/app/oracle/product/12.1.0.2.0/dbhome_1/dbs/initORCL_before.ora' from spfile;

File created.

SQL> alter system set db_unique_name='ORCL' scope=spfile;

System altered.

SQL> ALTER SYSTEM SET LOG_ARCHIVE_CONFIG='DG_CONFIG=(ORCL,ORCL_DR)' scope=both;

System altered.
```

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_DEST\_2='SERVICE=ORCL\_DR LGWR ASYNC  
VALID\_FOR=(ONLINE\_LOGFILES,PRIMARY\_ROLE) DB\_UNIQUE\_NAME=ORCL\_DR' scope=both;

System altered.

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_DEST\_STATE\_1=ENABLE scope=both;

System altered.

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_DEST\_STATE\_2=ENABLE scope=both;

System altered.

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_FORMAT='%t\_%s\_%r.dbf' SCOPE=SPFILE;

System altered.

SQL> ALTER SYSTEM SET LOG\_ARCHIVE\_MAX\_PROCESSES=3 scope=both;

System altered.

SQL> ALTER SYSTEM SET REMOTE\_LOGIN\_PASSWORDFILE=EXCLUSIVE SCOPE=SPFILE;

System altered.

```
SQL> ALTER SYSTEM SET fal_client=ORCLPR scope=both;
```

System altered.

```
SQL> ALTER SYSTEM SET fal_server=ORCL_DR scope=both;
```

System altered.

```
SQL> ALTER SYSTEM SET  
DB_FILE_NAME_CONVERT='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/', '/u01/app/ora  
cle/oradata/ORCLPR/16B447BB05687DB8E063060A14ACBD2D/', '/u01/app/oracle/oradata/ORCL_DR/DRPDB/'  
SCOPE=SPFILE;
```

System altered.

```
SQL> ALTER SYSTEM SET  
LOG_FILE_NAME_CONVERT='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/' SCOPE=SPFILE;
```

System altered.

```
SQL> ALTER SYSTEM SET STANDBY_FILE_MANAGEMENT=AUTO;
```

System altered.

```
oracle@pr:~  
SQL> show parameter pfile  


| NAME   | TYPE   | VALUE                                                            |
|--------|--------|------------------------------------------------------------------|
| spfile | string | /u01/app/oracle/product/19.0.0<br>/dbhome_1/dbs/spfileorclpr.ora |

  
SQL> create pfile='/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initORCLPR_before.ora' from spfile;  
File created.  
  
SQL> alter system set db_unique_name='ORCLPR' scope=spfile;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_CONFIG='DG_CONFIG=(ORCLPR,ORCL_DR)' scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_2='SERVICE=ORCL_DR LGWR ASYNC VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=ORCL_DR' scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_FORMAT='%t_%s_%r.dbf' SCOPE=SPFILE;  
System altered.  
  
SQL> ALTER SYSTEM SET LOG_ARCHIVE_MAX_PROCESSES=3 scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE SCOPE=SPFILE;  
System altered.  
  
SQL> ALTER SYSTEM SET fal_client=ORCLPR scope=both;  
System altered.  
  
SQL> ALTER SYSTEM SET fal_server=ORCL_DR scope=both;  
System altered.
```

SQL> create pfile='/u01/app/oracle/product/19.0.0/dbhome\_1/dbs/initORCLPR.ora' from spfile;

File created.

➤ Cat the init file after creating pfile from spfile:

```
oracle@pr:/u01/app/oracle/product/19.0.0/dbhome_1/dbs
[oracle@pr dbs]$ cat initORCLPR.ora
orclpr._data_transfer_cache_size=0
orclpr._db_cache_size=822083584
orclpr._inmemory_ext_roarea=0
orclpr._inmemory_ext_rwarea=0
orclpr._java_pool_size=0
orclpr._large_pool_size=16777216
orclpr._oracle_base='/u01/app/oracle'#ORACLE_BASE set from environment
orclpr._pga_aggregate_target=419430400
orclpr._sga_target=1241513984
orclpr._shared_io_pool_size=67108864
orclpr._shared_pool_size=318767104
orclpr._streams_pool_size=0
orclpr._unified_pga_pool_size=0
*.audit_file_dest='/u01/app/oracle/admin/orclpr/adump'
*.audit_trail='db'
*.compatible='19.0.0'
*.control_files='/u01/app/oracle/oradata/ORCLPR/controlfile/o1_mf_m2f8w73d_.ctl','/u01/app/oracle/fast_recovery_area/ORCLPR/controlfile/o1_mf_m2f8w74z_.ctl'
*.db_block_size=8192
*.db_create_file_dest='/u01/app/oracle/oradata'
*.db_file_name_convert='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/', '/u01/app/oracle/oradata/ORCLPR/16B447BB05687DB8E063060A14ACBD2D/', '/u01/app/oracle/oradata/ORCL_DR/DRPDB/'
*.db_name='orclpr'
*.db_recovery_file_dest='/u01/app/oracle/fast_recovery_area'
*.db_recovery_file_dest_size=5000m
*.db_unique_name='ORCLPR'
*.diagnostic_dest='/u01/app/oracle'
*.dispatchers='(PROTOCOL=TCP) (SERVICE=orclprXDB)'
*.enable_pluggable_database=true
*.fal_client='ORCLPR'
*.fal_server='ORCL_DR'
*.log_archive_config='DG_CONFIG=(ORCLPR,ORCL_DR)'
*.log_archive_dest_1='location=/u01/app/oracle/oradata/ORCLPR/archive/'
*.log_archive_dest_2='SERVICE=ORCL_DR LGWR ASYNC VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=ORCL_DR'
*.log_archive_dest_state_1='ENABLE'
*.log_archive_dest_state_2='ENABLE'
*.log_archive_format='%t_%s_%r.dbf'
*.log_archive_max_processes=3
*.log_file_name_convert='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/'
*.nls_language='AMERICAN'
*.nls_territory='AMERICA'
*.open_cursors=300
*.pga_aggregate_target=393m
*.processes=300
*.remote_login_passwordfile='EXCLUSIVE'
*.sga_target=1176m
*.standby_file_management='AUTO'
*.undo_tablespace='UNDOTBS1'
[oracle@pr dbs]$
```

## 6. Transfer SPFILE AND PASSWORD FILE TO STANDBY SIDE:

Copy the password file from the primary \$ORACLE\_HOME/dbs and rename it to the standby database name.

The username is required to be SYS and the password needs to be the same on the Primary and Standby.

The best practice for this is to copy the passwordfile as suggested.

The password file name must match the instance name/SID used at the standby site, not the DB\_NAME.

```
[oracle@pr dbs]$ scp initORCLPR.ora 172.20.10.9:/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initORCL_DR.ora
```

The authenticity of host '172.20.10.9 (172.20.10.9)' can't be established.

ECDSA key fingerprint is SHA256:VO98FP3v3Ti0CJ78Yj2EhEA7+3DPq+FhKePUwapEgRI.

ECDSA key fingerprint is MD5:0c:d8:ad:49:08:20:04:14:5c:16:0b:35:13:02:0f:7a.

Are you sure you want to continue connecting (yes/no)? yes

Warning: Permanently added '172.20.10.9' (ECDSA) to the list of known hosts.

oracle@172.20.10.9's password:

```
initORCLPR.ora 100% 1806 278.9KB/s
```

00:00

```
[oracle@pr dbs]$ scp orapworclpr 172.20.10.9:/u01/app/oracle/product/19.0.0/dbhome_1/dbs/orapwORCL_DR
```

oracle@172.20.10.9's password:

```
orapworclpr 100% 7680 728.9KB/s
```

00:00

```
[oracle@pr dbs]$
```

## 7. Prepare an Initialization Parameter File for the Standby Database on Standby:

Edit shown highlighted to set it up for the standby role

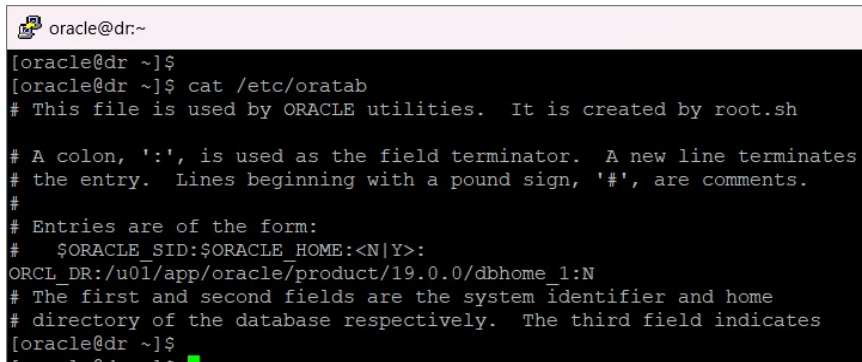
```
[oracle@pr dbs]$ cat initORCLPR.ora
orclpr.__data_transfer_cache_size=0
orclpr.__db_cache_size=822083584
orclpr.__inmemory_ext_roarea=0
orclpr.__inmemory_ext_rwarea=0
orclpr.__java_pool_size=0
orclpr.__large_pool_size=16777216
orclpr.__oracle_base='/u01/app/oracle'#ORACLE_BASE set from environment
orclpr.__pga_aggregate_target=419430400
orclpr.__sga_target=1241513984
orclpr.__shared_io_pool_size=67108864
orclpr.__shared_pool_size=318767104
orclpr.__streams_pool_size=0
orclpr.__unified_pga_pool_size=0
*.audit_file_dest='/u01/app/oracle/admin/ORCL_DR/adump'
*.audit_trail='db'
*.compatible='19.0.0'
*.control_files='/u01/app/oracle/oradata/ORCL_DR/controlfile/o1_mf_m2f8w73d_.ctl','/u01/app/oracle/fast_recovery_area/ORCL_DR/controlfile/o1_mf_m2f8w74z_.ctl'
*.db_block_size=8192
*.db_create_file_dest='/u01/app/oracle/oradata'
*.db_file_name_convert='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/', '/u01/app/oracle/oradata/ORCLPR/16B447BB05687DB8E063060A14ACBD2D/', '/u01/app/oracle/oradata/ORCL_DR/DRPDB/'
*.db_name='orclpr'
*.db_recovery_file_dest='/u01/app/oracle/fast_recovery_area'
*.db_recovery_file_dest_size=5000m
*.db_unique_name='ORCL_DR'
*.diagnostic_dest='/u01/app/oracle'
*.dispatchers='(PROTOCOL=TCP) (SERVICE=ORCL_DRXDB)'
*.enable_pluggable_database=true
*.fal_client='ORCLPR'
*.fal_server='ORCL_DR'
*.log_archive_config='DG_CONFIG=(ORCLPR,ORCL_DR)'
*.log_archive_dest_1='location=/u01/app/oracle/oradata/ORCL_DR/archive'
*.log_archive_dest_2='SERVICE=ORCL_DR LGWR ASYNC VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=ORCL_DR'
*.log_archive_dest_state_1='ENABLE'
*.log_archive_dest_state_2='ENABLE'
*.log_archive_format='%t_%s_%r.dbf'
*.log_archive_max_processes=3
*.log_file_name_convert='/u01/app/oracle/oradata/ORCLPR/', '/u01/app/oracle/oradata/ORCL_DR/'
*.nls_language='AMERICAN'
*.nls_territory='AMERICA'
*.open_cursors=300
*.pga_aggregate_target=393m
*.processes=300
*.remote_login_passwordfile='EXCLUSIVE'
*.sga_target=1176m
*.standby_file_management='AUTO'
*.undo_tablespace='UNDOTBS1'
[oracle@pr dbs]$
```

## 8. Create required directories on Standby:

```
[oracle@dr ~]$ mkdir -p /u01/app/oracle/admin/ORCL_DR/adump
[oracle@dr ~]$ mkdir -p /u01/app/oracle/oradata/ORCL_DR
[oracle@dr ~]$ mkdir -p /u01/app/oracle/oradata/ORCL_DR/archive
[oracle@dr ~]$ mkdir -p /u01/app/oracle/oradata/ORCL_DR/DRPDB
```

## 9. Add below entry in ORATAB on Standby:

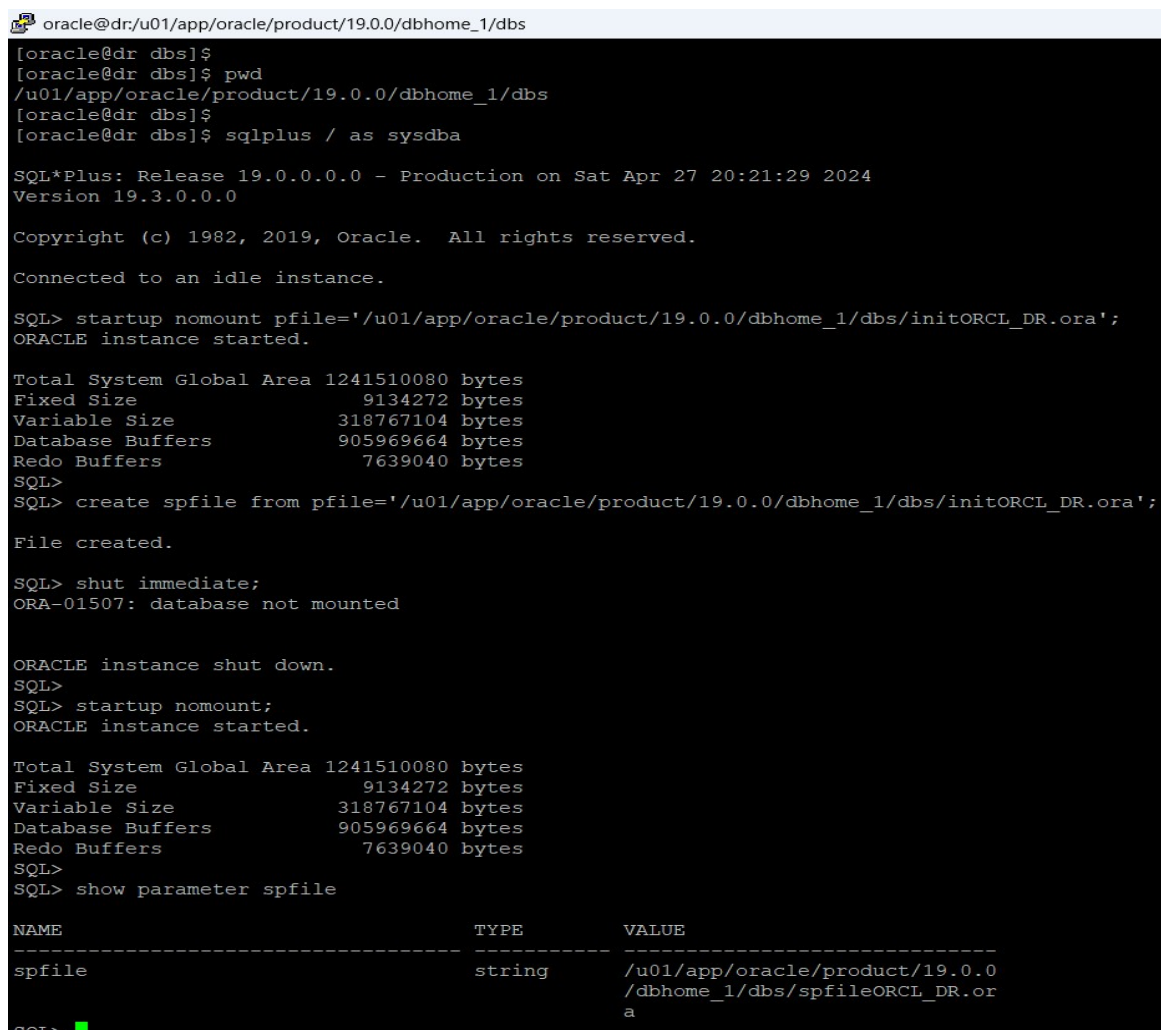
```
[oracle@dr ~]$ cat /etc/oratab
ORCL_DR:/u01/app/oracle/product/19.0.0/dbhome_1:N
[oracle@dr ~]$
```



```
oracle@dr:~
[oracle@dr ~]$
[oracle@dr ~]$ cat /etc/oratab
# This file is used by ORACLE utilities.  It is created by root.sh

# A colon, ':', is used as the field terminator.  A new line terminates
# the entry.  Lines beginning with a pound sign, '#', are comments.
#
# Entries are of the form:
#   $ORACLE_SID:$ORACLE_HOME:<N|Y>:
ORCL_DR:/u01/app/oracle/product/19.0.0/dbhome_1:N
# The first and second fields are the system identifier and home
# directory of the database respectively.  The third field indicates
[oracle@dr ~]$
```

## 10. Create spfile from pfile on Standby:



```
oracle@dr:/u01/app/oracle/product/19.0.0/dbhome_1/dbs
[oracle@dr dbs]$
[oracle@dr dbs]$ pwd
/u01/app/oracle/product/19.0.0/dbhome_1/dbs
[oracle@dr dbs]$
[oracle@dr dbs]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:21:29 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to an idle instance.

SQL> startup nomount pfile='/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initORCL_DR.ora';
ORACLE instance started.

Total System Global Area 1241510080 bytes
Fixed Size 9134272 bytes
Variable Size 318767104 bytes
Database Buffers 905969664 bytes
Redo Buffers 7639040 bytes
SQL>
SQL> create spfile from pfile='/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initORCL_DR.ora';
File created.

SQL> shut immediate;
ORA-01507: database not mounted

ORACLE instance shut down.
SQL>
SQL> startup nomount;
ORACLE instance started.

Total System Global Area 1241510080 bytes
Fixed Size 9134272 bytes
Variable Size 318767104 bytes
Database Buffers 905969664 bytes
Redo Buffers 7639040 bytes
SQL>
SQL> show parameter spfile

NAME                                 TYPE        VALUE
-----
spfile                               string      /u01/app/oracle/product/19.0.0
                                         /dbhome_1/dbs/spfileORCL_DR.or
                                         a
SQL>
```

## 11. Configure TNS Entries on Primary:

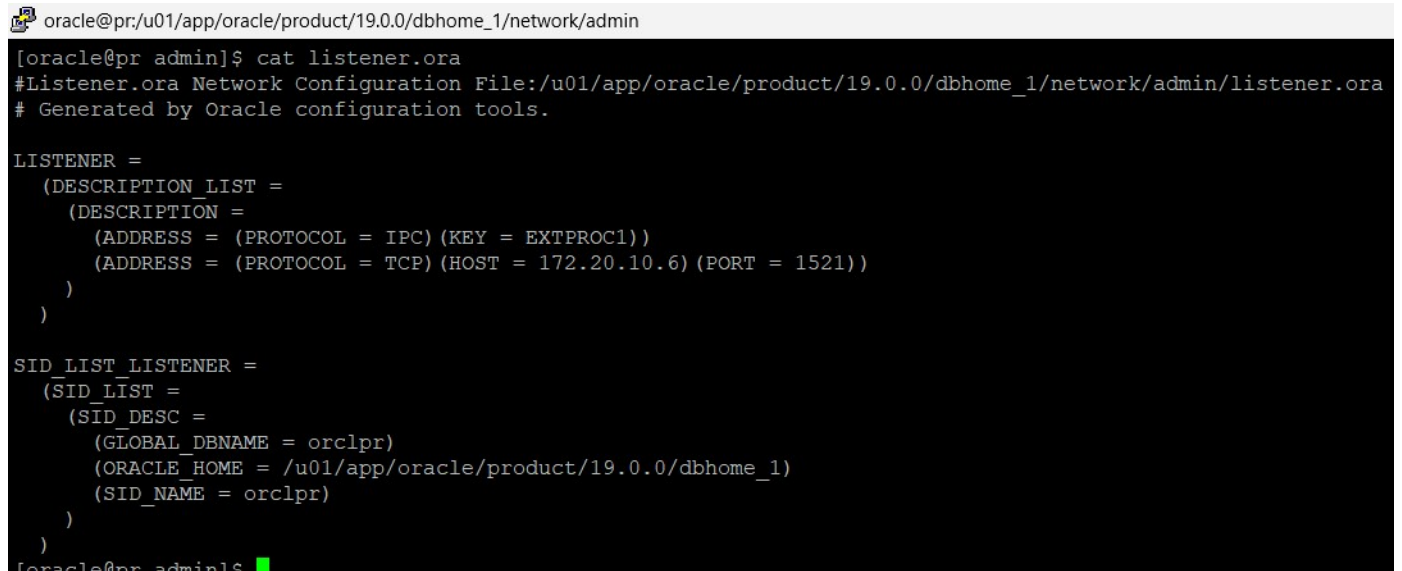
```
[oracle@pr admin]$ cat listener.ora
```

```
#Listener.ora Network Configuration File:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
```

```
# Generated by Oracle configuration tools.
```

```
LISTENER =  
  (DESCRIPTION_LIST =  
    (DESCRIPTION =  
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1))  
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))  
    )  
  )
```

```
SID_LIST_LISTENER =  
  (SID_LIST =  
    (SID_DESC =  
      (GLOBAL_DBNAME = orclpr)  
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)  
      (SID_NAME = orclpr)  
    )  
  )
```



The screenshot shows a terminal window with the title bar "oracle@pr:/u01/app/oracle/product/19.0.0/dbhome\_1/network/admin". The terminal displays the command `[oracle@pr admin]$ cat listener.ora` and the output of the file, which is the same as the text provided in the previous blocks. The terminal background is black with white text.

```
[oracle@pr admin]$ cat tnsnames.ora
```

```
# tnsnames.ora Network Configuration File:
```

```
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora
```

```
# Generated by Oracle configuration tools.
```

```
LISTENER = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))
```

```
ORCLPR =  
  (DESCRIPTION =  
    (ADDRESS_LIST =  
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))  
    )  
    (CONNECT_DATA =  
      (SERVICE_NAME = orclpr)  
    )  
  )
```

```
)

PRODPDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRODPDB)
    )
  )
)
```

```
ORCL_DR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = ORCL_DR)
    )
  )
)
```

```
[oracle@pr admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

LISTENER = (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.6) (PORT = 1521))

ORCLPR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.6) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = orclpr)
    )
  )

PRODPDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.6) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRODPDB)
    )
  )

ORCL_DR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.9) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = ORCL_DR)
    )
  )
[oracle@pr admin]$
```

➤ Check the status of listener: Command - lsnrctl status

```
oracle@pr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@pr admin]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:42:28

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=EXTPROC)))
STATUS of the LISTENER
-----
Alias                     listener
Version                  TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date               25-APR-2024 17:37:54
Uptime                   0 days 2 hr. 4 min. 34 sec
Trace Level              off
Security                 ON: Local OS Authentication
SNMP                     OFF
Listener Parameter File  /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File        /u01/app/oracle/diag/tnslsnr/pr/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=172.20.10.6)(PORT=1521)))
Services Summary...
Service "16b447bb05687db8e063060a14acbd2d" has 1 instance(s).
  Instance "orclpr", status READY, has 1 handler(s) for this service...
Service "orclpr" has 2 instance(s).
  Instance "orclpr", status UNKNOWN, has 1 handler(s) for this service...
  Instance "orclpr", status READY, has 1 handler(s) for this service...
Service "orclprXDB" has 1 instance(s).
  Instance "orclpr", status READY, has 1 handler(s) for this service...
Service "prodpdb" has 1 instance(s).
  Instance "orclpr", status READY, has 1 handler(s) for this service...
The command completed successfully

[oracle@pr admin]$
```

➤ Now to check the tnsping on PR node of both server:

```
oracle@pr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@pr admin]$ tnsping orclpr

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:44:01

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = orclpr)))
OK (10 msec)
[oracle@pr admin]$ tnsping ORCL_DR

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:44:05

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = ORCL_DR)))
OK (10 msec)
```

## 12. Configure TNS Entries on Standby:

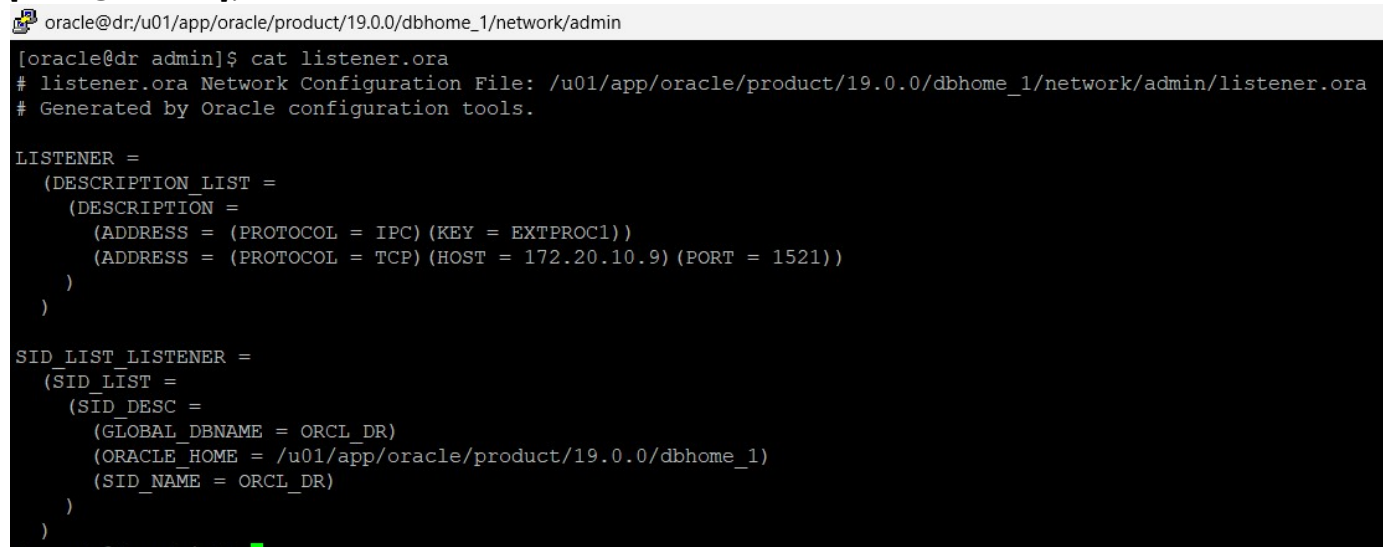
```
[oracle@dr admin]$ cat listener.ora
```

```
# listener.ora Network Configuration File: /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
# Generated by Oracle configuration tools.
```

```
LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1))
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))
    )
  )
```

```
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = ORCL_DR)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = ORCL_DR)
    )
  )
```

```
[oracle@dr admin]$
```

A screenshot of a terminal window with a dark background. The prompt is [oracle@dr:u01/app/oracle/product/19.0.0/dbhome\_1/network/admin]. The user has entered the command cat listener.ora. The output shows the contents of the listener.ora file, which includes the LISTENER and SID\_LIST\_LISTENER sections. The LISTENER section defines a description list with two addresses: one for IPC (KEY = EXTPROC1) and one for TCP (HOST = 172.20.10.9, PORT = 1521). The SID\_LIST\_LISTENER section defines a list of SID descriptors, with one descriptor for ORCL\_DR, specifying its global database name, Oracle home, and SID name.

```
[oracle@dr:u01/app/oracle/product/19.0.0/dbhome_1/network/admin]
[oracle@dr admin]$ cat listener.ora
# listener.ora Network Configuration File: /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1))
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))
    )
  )

SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = ORCL_DR)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = ORCL_DR)
    )
  )
```

```
[oracle@dr admin]$ cat tnsnames.ora
```

```
# tnsnames.ora Network Configuration File:
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.
```

```
LISTENER = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))
```

```
ORCL_DR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = ORCL_DR)
    )
  )
```

```

ORCLPR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = orclpr)
    )
  )

```

```

PRODPDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRODPDB)
    )
  )

```

 oracle@dr:/u01/app/oracle/product/19.0.0/dbhome\_1/network/admin

```

[oracle@dr admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

LISTENER = (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.9) (PORT = 1521))

ORCL_DR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.9) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = ORCL_DR)
    )
  )

ORCLPR =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.6) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = orclpr)
    )
  )

PRODPDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = 172.20.10.6) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PRODPDB)
    )
  )

```

- After the tns & listener entries connect to DB and do below steps, to get the listener status:

```
oracle@dr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@dr admin]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Apr 25 19:48:20 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> shut immediate;
ORA-01507: database not mounted

ORACLE instance shut down.
SQL> startup nomount;
ORACLE instance started.

Total System Global Area 1241510080 bytes
Fixed Size 9134272 bytes
Variable Size 318767104 bytes
Database Buffers 905969664 bytes
Redo Buffers 7639040 bytes
SQL>
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@dr admin]$
```

- Check the listener status on DR:

```
oracle@dr:/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@dr admin]$ lsnrctl status listener

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:50:25

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC) (KEY=EXTPROC)))
STATUS of the LISTENER
-----
Alias                listener
Version              TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date           25-APR-2024 19:50:13
Uptime               0 days 0 hr. 0 min. 12 sec
Trace Level          off
Security             ON: Local OS Authentication
SNMP                 OFF
Listener Parameter File /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File    /u01/app/oracle/diag/tnslsnr/dr/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc) (KEY=EXTPROC)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=172.20.10.9) (PORT=1521)))
Services Summary...
Service "ORCL_DR" has 2 instance(s).
  Instance "ORCL_DR", status UNKNOWN, has 1 handler(s) for this service...
  Instance "orcl_dr", status BLOCKED, has 1 handler(s) for this service...
The command completed successfully
[oracle@dr admin]$
```

### ➤ Now to check the tnsping on DR node of both server:

```
oracle@dr/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@dr admin]$ tnsping ORCL_DR

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:51:04

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.9)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = ORCL_DR)))
OK (10 msec)
[oracle@dr admin]$ tnsping orclpr

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 25-APR-2024 19:51:11

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 172.20.10.6)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = orclpr)))
OK (80 msec)
[oracle@dr admin]$
```

### 13. Verify connection 'AS SYSDBA' from Primary:

[oracle@pr ~]\$ sqlplus sys/Password#1@ORCLPR as sysdba

[oracle@pr ~]\$ sqlplus sys/Password#1@ORCL\_DR as sysdba

```
oracle@pr:~
[oracle@pr ~]$
[oracle@pr ~]$ sqlplus sys/Password#1@ORCLPR as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:33:41 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@pr ~]$
[oracle@pr ~]$ sqlplus sys/Password#1@ORCL_DR as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:33:58 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@pr ~]$
```

## 14. Create Standby Database using duplicate command:

On Primary:

```
[oracle@pr ~]$ rman target sys/Password#1@ORCLPR auxiliary sys/Password#1@ORCL_DR
```

Recovery Manager: Release 19.0.0.0.0 - Production on Sat Apr 27 20:35:44 2024  
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.

connected to target database: ORCLPR (DBID=829140703)

connected to auxiliary database: ORCLPR (not mounted)

```
RMAN> duplicate target database for standby from active database nofilenamecheck;
```

Starting Duplicate Db at 27-APR-24  
using target database control file instead of recovery catalog  
allocated channel: ORA\_AUX\_DISK\_1  
channel ORA\_AUX\_DISK\_1: SID=22 device type=DISK

contents of Memory Script:

```
{  
  backup as copy reuse  
  passwordfile auxiliary format '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/orapwORCL_DR' ;  
}
```

executing Memory Script

Starting backup at 27-APR-24  
allocated channel: ORA\_DISK\_1  
channel ORA\_DISK\_1: SID=270 device type=DISK  
Finished backup at 27-APR-24

contents of Memory Script:

```
{  
  sql clone "alter system set control_files =  
  '/u01/app/oracle/oradata/ORCL_DR/controlfile/o1_mf_m2f8w73d_.ctl',  
  '/u01/app/oracle/fast_recovery_area/ORCL_DR/controlfile/o1_mf_m2f8w74z_.ctl' comment=  
  'Set by RMAN' scope=spfile";  
  restore clone from service 'ORCLPR' standby controlfile;  
}
```

executing Memory Script

sql statement: alter system set control\_files = '/u01/app/oracle/oradata/ORCL\_DR/controlfile/o1\_mf\_m2f8w73d\_.ctl',  
'/u01/app/oracle/fast\_recovery\_area/ORCL\_DR/controlfile/o1\_mf\_m2f8w74z\_.ctl' comment= 'Set by RMAN' scope=spfile

Starting restore at 27-APR-24  
using channel ORA\_AUX\_DISK\_1

channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: restoring control file  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:04  
output file name=/u01/app/oracle/oradata/ORCL\_DR/controlfile/o1\_mf\_m2f8w73d\_.ctl  
output file name=/u01/app/oracle/fast\_recovery\_area/ORCL\_DR/controlfile/o1\_mf\_m2f8w74z\_.ctl  
Finished restore at 27-APR-24

contents of Memory Script:

```
{  
  sql clone 'alter database mount standby database';  
}
```

executing Memory Script

sql statement: alter database mount standby database

contents of Memory Script:

```

{
  set newname for tempfile 1 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_temp_m2f8wmd6_.tmp";
  set newname for tempfile 2 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/temp012024-04-22_23-25-22-574-PM.dbf";
  set newname for tempfile 3 to
"/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_temp_m2f9pz2m_.dbf";
  switch clone tempfile all;
  set newname for datafile 1 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_system_m2f8s6wj_.dbf";
  set newname for datafile 3 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_sysaux_m2f8tb9z_.dbf";
  set newname for datafile 4 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_undotbs1_m2f8tsfr_.dbf";
  set newname for datafile 5 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_system_m2f937l9_.dbf";
  set newname for datafile 6 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_sysaux_m2f937lr_.dbf";
  set newname for datafile 7 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_users_m2f8ttjg_.dbf";
  set newname for datafile 8 to
"/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_undotbs1_m2f937ly_.dbf";
  set newname for datafile 9 to
"/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_system_m2f9pz27_.dbf";
  set newname for datafile 10 to
"/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_sysaux_m2f9pz2k_.dbf";
  set newname for datafile 11 to
"/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_undotbs1_m2f9pz2l_.dbf";
  set newname for datafile 12 to
"/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_users_m2f9qdwy_.dbf";
  restore
  from nonparse from service
'ORCLPR' clone database
;
  sql 'alter system archive log current';
}
executing Memory Script
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME

```

renamed tempfile 1 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_temp\_m2f8wmd6\_.tmp in control file  
renamed tempfile 2 to /u01/app/oracle/oradata/ORCL\_DR/datafile/temp012024-04-22\_23-25-22-574-PM.dbf in control file  
renamed tempfile 3 to  
/u01/app/oracle/oradata/ORCL\_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1\_mf\_temp\_m2f9pz2m\_.dbf in control file

```

executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME

```

Starting restore at 27-APR-24  
using channel ORA\_AUX\_DISK\_1

```

channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service ORCLPR
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_system_m2f8s6wj_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:45
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service ORCLPR
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

```

channel ORA\_AUX\_DISK\_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_sysaux\_m2f8tb9z\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:25  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_undotbs1\_m2f8tsfr\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:03  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_system\_m2f937l9\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:16  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00006 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_sysaux\_m2f937lr\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:25  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_users\_m2f8ttjg\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:02  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_undotbs1\_m2f937ly\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:07  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00009 to  
/u01/app/oracle/oradata/ORCL\_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1\_mf\_system\_m2f9pz27\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:16  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00010 to  
/u01/app/oracle/oradata/ORCL\_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1\_mf\_sysaux\_m2f9pz2k\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:15  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00011 to  
/u01/app/oracle/oradata/ORCL\_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1\_mf\_undotbs1\_m2f9pz2l\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:04  
channel ORA\_AUX\_DISK\_1: starting datafile backup set restore  
channel ORA\_AUX\_DISK\_1: using network backup set from service ORCLPR  
channel ORA\_AUX\_DISK\_1: specifying datafile(s) to restore from backup set  
channel ORA\_AUX\_DISK\_1: restoring datafile 00012 to  
/u01/app/oracle/oradata/ORCL\_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1\_mf\_users\_m2f9qdwy\_.dbf  
channel ORA\_AUX\_DISK\_1: restore complete, elapsed time: 00:00:01  
Finished restore at 27-APR-24

sql statement: alter system archive log current

contents of Memory Script:

```
{
  switch clone datafile all;
}
```

executing Memory Script

datafile 1 switched to datafile copy

input datafile copy RECID=4 STAMP=1167424760 file name=/u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_system\_m2t53zfr\_.dbf

datafile 3 switched to datafile copy

input datafile copy RECID=5 STAMP=1167424760 file name=/u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_sysaux\_m2t55f6g\_.dbf

datafile 4 switched to datafile copy

input datafile copy RECID=6 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL\_DR/datafile/o1\_mf\_undotbs1\_m2t566vf\_.dbf

datafile 5 switched to datafile copy

```

input datafile copy RECID=7 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_system_m2t56bkw_.dbf
datafile 6 switched to datafile copy
input datafile copy RECID=8 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_sysaux_m2t56syv_.dbf
datafile 7 switched to datafile copy
input datafile copy RECID=9 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_users_m2t57mlg_.dbf
datafile 8 switched to datafile copy
input datafile copy RECID=10 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_undotbs1_m2t57nxi_.dbf
datafile 9 switched to datafile copy
input datafile copy RECID=11 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_system_m2t57wlk_.dbf
datafile 10 switched to datafile copy
input datafile copy RECID=12 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_sysaux_m2t58d30_.dbf
datafile 11 switched to datafile copy
input datafile copy RECID=13 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_undotbs1_m2t58vpl_.dbf
datafile 12 switched to datafile copy
input datafile copy RECID=14 STAMP=1167424761 file
name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_users_m2t58z08_.dbf
Finished Duplicate Db at 27-APR-24

```

RMAN> exit

Recovery Manager complete.

[oracle@pr ~]\$

```

oracle@pr:~
[oracle@pr ~]$ rman target sys/Password#1@ORCLPR auxiliary sys/Password#1@ORCL_DR

Recovery Manager: Release 19.0.0.0.0 - Production on Sat Apr 27 20:35:44 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.

connected to target database: ORCLPR (DBID=829140703)
connected to auxiliary database: ORCLPR (not mounted)

RMAN> duplicate target database for standby from active database nofilenamecheck;

Starting Duplicate Db at 27-APR-24
using target database control file instead of recovery catalog
allocated channel: ORA_AUX_DISK_1
channel ORA_AUX_DISK_1: SID=22 device type=DISK

contents of Memory Script:
{
  backup as copy reuse
    passwordfile auxiliary format '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/orapwORCL_DR' ;
}
executing Memory Script

Starting backup at 27-APR-24
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=270 device type=DISK
Finished backup at 27-APR-24

contents of Memory Script:
{
  sql clone "alter system set control_files =
    '/u01/app/oracle/oradata/ORCL_DR/controlfile/o1_mf_m2f8w73d_.ctl', '/u01/app/oracle/fast_recovery_area/ORCL_DR/controlfile/o1_mf_m2f8w74z_.ctl' comment=
    'Set by RMAN'" scope=spfile";
  restore clone from service 'ORCLPR' standby controlfile;
}
executing Memory Script

sql statement: alter system set control_files = '/u01/app/oracle/oradata/ORCL_DR/controlfile/o1_mf_m2f8w73d_.ctl', '/u01/app/oracle/fast_recovery_area/ORCL_DR/controlfile/o1_mf_m2f8w74z_.ctl' comment= 'Set by RMAN' scope=spfile

Starting restore at 27-APR-24
using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service ORCLPR
channel ORA_AUX_DISK_1: restoring control file
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:04
output file name=/u01/app/oracle/oradata/ORCL_DR/controlfile/o1_mf_m2f8w73d_.ctl

```

```

oracle@pr~
channel ORA_AUX_DISK_1: using network backup set from service ORCLPR
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00011 to /u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_undotbs1_m2f9pz21_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:04
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service ORCLPR
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00012 to /u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_users_m2f9gdwy_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
Finished restore at 27-APR-24

sql statement: alter system archive log current

contents of Memory Script:
{
  switch clone datafile all;
}
executing Memory Script

datafile 1 switched to datafile copy
input datafile copy RECID=4 STAMP=1167424760 file name=/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_system_m2t53zfr_.dbf
datafile 3 switched to datafile copy
input datafile copy RECID=5 STAMP=1167424760 file name=/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_sysaux_m2t55f6g_.dbf
datafile 4 switched to datafile copy
input datafile copy RECID=6 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_undotbs1_m2t566vf_.dbf
datafile 5 switched to datafile copy
input datafile copy RECID=7 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_system_m2t56bkw_.dbf
datafile 6 switched to datafile copy
input datafile copy RECID=8 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_sysaux_m2t56syv_.dbf
datafile 7 switched to datafile copy
input datafile copy RECID=9 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/datafile/o1_mf_users_m2t57mlg_.dbf
datafile 8 switched to datafile copy
input datafile copy RECID=10 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B42434516A766CE063060A14AC3A5C/datafile/o1_mf_undotbs1_m2t57nxi_.dbf
datafile 9 switched to datafile copy
input datafile copy RECID=11 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_system_m2t57w1k_.dbf
datafile 10 switched to datafile copy
input datafile copy RECID=12 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_sysaux_m2t58d30_.dbf
datafile 11 switched to datafile copy
input datafile copy RECID=13 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_undotbs1_m2t58vpl_.dbf
datafile 12 switched to datafile copy
input datafile copy RECID=14 STAMP=1167424761 file name=/u01/app/oracle/oradata/ORCL_DR/16B447BB05687DB8E063060A14ACBD2D/datafile/o1_mf_users_m2t58z08_.dbf
Finished Duplicate Db at 27-APR-24

RMAN> exit

Recovery Manager complete.
[oracle@pr ~]$

```

## 15. Enable MRP on Standby: [On DR]

```

[oracle@dr ~]$ . oraenv
ORACLE_SID = [ORCL_DR] ?
The Oracle base remains unchanged with value /u01/app/oracle
[oracle@dr ~]$
[oracle@dr ~]$ sqlplus / as sysdba

```

SQL\*Plus: Release 12.1.0.2.0 Production on Thu Apr 18 19:30:31 2024  
 Copyright (c) 1982, 2014, Oracle. All rights reserved.  
 Connected to:  
 Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 - 64bit Production  
 With the Partitioning, OLAP, Advanced Analytics and Real Application Testing options

```
SQL> select name,open_mode,database_role,cdb from v$database;
```

| NAME | OPEN_MODE | DATABASE_ROLE    | CDB |
|------|-----------|------------------|-----|
| ORCL | MOUNTED   | PHYSICAL STANDBY | YES |

```
SQL> alter database open;
```

Database altered.

```

SQL> !ps -ef | grep mrp
oracle 8221 6347 0 19:33 pts/0 00:00:00 /bin/bash -c ps -ef | grep mrp
oracle 8223 8221 0 19:33 pts/0 00:00:00 grep mrp

```

```
SQL> alter database recover managed standby database disconnect from session;
```

Database altered.

```
SQL> select name,open_mode,database_role,cdb from v$database;
```

| NAME | OPEN_MODE            | DATABASE_ROLE    | CDB |
|------|----------------------|------------------|-----|
| ORCL | READ ONLY WITH APPLY | PHYSICAL STANDBY | YES |

```
oracle@dr:~
root@dr ~]# su - oracle
Last login: Sat Apr 27 19:52:02 IST 2024 on pts/1
oracle@dr ~]$
oracle@dr ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:45:14 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> select name,open_mode,database_role,cdb from v$database;

NAME          OPEN_MODE          DATABASE_ROLE      CDB
-----
ORCLPR        MOUNTED            PHYSICAL STANDBY   YES

SQL> alter database open;

Database altered.

SQL> !ps -ef | grep mrp
oracle      9487   9405   0 20:45 pts/1    00:00:00 /bin/bash -c ps -ef | grep mrp
oracle      9489   9487   0 20:45 pts/1    00:00:00 grep mrp

SQL> alter database recover managed standby database disconnect from session;

Database altered.

SQL> select name,open_mode,database_role,cdb from v$database;

NAME          OPEN_MODE          DATABASE_ROLE      CDB
-----
ORCLPR        READ ONLY WITH APPLY PHYSICAL STANDBY   YES

SQL> col name for a30
SQL> select name,open_mode,con_id,dbid from v$containers;

NAME          OPEN_MODE          CON_ID      DBID
-----
CDB$ROOT      READ ONLY          1  829140703
CDB$SEED      READ ONLY          2  3650434618
PRODPDB       MOUNTED            3  3936623554

SQL>
```

## 16. Open PDB in READ-ONLY Mode:

```
SQL> alter pluggable database all open read only;
```

Pluggable database altered.

```
SQL> select name,open_mode,con_id,dbid from v$containers;
```

| NAME      | OPEN_MODE | CON_ID | DBID       |
|-----------|-----------|--------|------------|
| CDB\$ROOT | READ ONLY | 1      | 1694349193 |
| PDB\$SEED | READ ONLY | 2      | 3387613093 |
| PDBORCL   | READ ONLY | 3      | 4152619178 |

```
SQL> alter pluggable database all open read only;

Pluggable database altered.

SQL> select name,open_mode,con_id,dbid from v$containers;

NAME          OPEN_MODE          CON_ID      DBID
-----
CDB$ROOT      READ ONLY          1  829140703
PDB$SEED      READ ONLY          2  3650434618
PRODPDB       READ ONLY          3  3936623554

SQL>
```

## 17. Verify the log shipping Is Performing Properly:

On Primary

```
[oracle@pr ~]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:59:03 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

Connected to:

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> show con_name
```

```
CON_NAME
```

```
-----  
CDB$ROOT
```

```
SQL>
```

```
SQL> show parameters db_unique_name;
```

| NAME           | TYPE   | VALUE  |
|----------------|--------|--------|
| db_unique_name | string | ORCLPR |

```
SQL>
```

```
SQL> SELECT SEQUENCE#, FIRST_TIME, NEXT_TIME FROM V$ARCHIVED_LOG ORDER BY SEQUENCE#;
```

```
SEQUENCE# FIRST_TIM NEXT_TIME
```

```
-----  
9 24-APR-24 25-APR-24  
10 25-APR-24 25-APR-24  
11 25-APR-24 25-APR-24  
12 25-APR-24 25-APR-24  
13 25-APR-24 25-APR-24  
14 25-APR-24 25-APR-24  
14 25-APR-24 25-APR-24  
15 25-APR-24 27-APR-24  
16 27-APR-24 27-APR-24  
17 27-APR-24 27-APR-24  
18 27-APR-24 27-APR-24
```

```
SEQUENCE# FIRST_TIM NEXT_TIME
```

```
-----  
19 27-APR-24 27-APR-24  
19 27-APR-24 27-APR-24
```

```
13 rows selected.
```

```

oracle@pr:~
[oracle@pr ~]$
[oracle@pr ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 20:59:03 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> show con_name

CON_NAME
-----
CDB$ROOT
SQL>
SQL> show parameters db_unique_name;

NAME                                TYPE        VALUE
-----
db_unique_name                      string      ORCLPR
SQL>
SQL> SELECT SEQUENCE#, FIRST_TIME, NEXT_TIME FROM V$ARCHIVED_LOG ORDER BY SEQUENCE#;

SEQUENCE# FIRST_TIM NEXT_TIME
-----
          9 24-APR-24 25-APR-24
         10 25-APR-24 25-APR-24
         11 25-APR-24 25-APR-24
         12 25-APR-24 25-APR-24
         13 25-APR-24 25-APR-24
         14 25-APR-24 25-APR-24
         14 25-APR-24 25-APR-24
         15 25-APR-24 27-APR-24
         16 27-APR-24 27-APR-24
         17 27-APR-24 27-APR-24
         18 27-APR-24 27-APR-24

SEQUENCE# FIRST_TIM NEXT_TIME
-----
         19 27-APR-24 27-APR-24
         19 27-APR-24 27-APR-24

13 rows selected.

```

SQL> alter system switch logfile;

System altered.

SQL> /

System altered.

SQL> /

System altered.

SQL> select max(sequence#) from v\$archived\_log where archived='YES';

MAX(SEQUENCE#)

-----  
22 <-----

SQL>

```

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
SQL> alter system switch logfile;

System altered.

SQL> /

System altered.

SQL> /

System altered.

SQL> select max(sequence#) from v$archived_log where archived='YES';

MAX(SEQUENCE#)
-----
                22

```

## On Standby:

```
SQL> SELECT SEQUENCE#, FIRST_TIME, NEXT_TIME FROM V$ARCHIVED_LOG ORDER BY SEQUENCE#;
```

```

SEQUENCE# FIRST_TIM NEXT_TIME
-----
19 27-APR-24 27-APR-24
20 27-APR-24 27-APR-24
21 27-APR-24 27-APR-24
22 27-APR-24 27-APR-24

```

```
SQL> select max(sequence#) from v$archived_log where applied='YES';
```

```

MAX(SEQUENCE#)
-----
                22  <----

```

```
SQL>
```

```
SQL>
```

```
SQL> SELECT SEQUENCE#, FIRST_TIME, NEXT_TIME FROM V$ARCHIVED_LOG ORDER BY SEQUENCE#;
```

```

SEQUENCE# FIRST_TIM NEXT_TIME
-----
19 27-APR-24 27-APR-24
20 27-APR-24 27-APR-24
21 27-APR-24 27-APR-24
22 27-APR-24 27-APR-24

```

```
SQL> select max(sequence#) from v$archived_log where applied='YES';
```

```

MAX(SEQUENCE#)
-----
                22

```

```
SQL>
```

➤ MRP process where status = Applying logs:

```
oracle@dr:~/scripts
[oracle@dr scripts]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Apr 27 21:16:18 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> col group# for a10
SQL> select PROCESS,STATUS,GROUP#,THREAD#,SEQUENCE#,BLOCK#,DELAY_MINS from v$managed_standby;

PROCESS      STATUS      GROUP#      THREAD#  SEQUENCE#    BLOCK#  DELAY_MINS
-----
ARCH         CLOSING      4             1         21           1         0
DGRD         ALLOCATED   N/A            0          0           0         0
DGRD         ALLOCATED   N/A            0          0           0         0
ARCH         CLOSING      5             1         22           1         0
ARCH         CONNECTED   N/A            0          0           0         0
RFS          IDLE        N/A            1          0           0         0
RFS          IDLE        2             1         23          435         0
RFS          IDLE        N/A            0          0           0         0
MRP0         APPLYING_LOG N/A            1         23          435         0

9 rows selected.

SQL> /

PROCESS      STATUS      GROUP#      THREAD#  SEQUENCE#    BLOCK#  DELAY_MINS
-----
ARCH         CLOSING      4             1         21           1         0
DGRD         ALLOCATED   N/A            0          0           0         0
DGRD         ALLOCATED   N/A            0          0           0         0
ARCH         CLOSING      5             1         22           1         0
ARCH         CONNECTED   N/A            0          0           0         0
RFS          IDLE        N/A            1          0           0         0
RFS          IDLE        2             1         23          438         0
RFS          IDLE        N/A            0          0           0         0
MRP0         APPLYING_LOG N/A            1         23          438         0

9 rows selected.
```

## 18. LETS CHECK THAT PRIMARY AND STANDBY DBs ARE IN SYNC:

At Primary make few switches and check current log sequence #23.

PRIMARY:

```
SQL> alter system switch all logfile;

System altered.

SQL> @sync_arch

  THREAD# Last Sequence Received Last Sequence Applied Difference
-----
        1             23             23             0
        1             23             23             0
```

STANDBY:

```
SQL> @sync_arch

  THREAD# Last Sequence Received Last Sequence Applied Difference
-----
        1             23             23             0

SQL>
```

On Standby site we can see that last sequence # 23 is applied on standby

➤ **Query to check sync status:**

```
set lines 500 pages 500
SELECT ARCH.THREAD# , ARCH.SEQUENCE# "Last Sequence Received", APPL.SEQUENCE# "Last Sequence Applied",
(ARCH.SEQUENCE# - APPL.SEQUENCE#) "Difference"
FROM (SELECT THREAD# ,SEQUENCE# FROM V$ARCHIVED_LOG WHERE (THREAD#,FIRST_TIME ) IN (SELECT
THREAD#,MAX(FIRST_TIME) FROM V$ARCHIVED_LOG GROUP BY THREAD#)) ARCH,
(SELECT THREAD# ,SEQUENCE# FROM V$LOG_HISTORY WHERE (THREAD#,FIRST_TIME ) IN (SELECT
THREAD#,MAX(FIRST_TIME) FROM V$LOG_HISTORY GROUP BY THREAD#)) APPL WHERE ARCH.THREAD# = APPL.THREAD#
ORDER BY 1;
```

---

## 19. Switchover status from Primary database:

PRIMARY:

```
SQL> set lines 400 pages 400
SQL> select NAME,OPEN_MODE,DATABASE_ROLE,DB_UNIQUE_NAME,SWITCHOVER_STATUS from v$database;
```

| NAME   | OPEN_MODE  | DATABASE_ROLE | DB_UNIQUE_NAME | SWITCHOVER_STATUS |
|--------|------------|---------------|----------------|-------------------|
| ORCLPR | READ WRITE | PRIMARY       | ORCLPR         | TO STANDBY        |

STANDBY:

```
SQL> set lines 400 pages 400
SQL> select NAME,OPEN_MODE,DATABASE_ROLE,DB_UNIQUE_NAME,SWITCHOVER_STATUS from v$database;
```

| NAME   | OPEN_MODE            | DATABASE_ROLE    | DB_UNIQUE_NAME | SWITCHOVER_STATUS |
|--------|----------------------|------------------|----------------|-------------------|
| ORCLPR | READ ONLY WITH APPLY | PHYSICAL STANDBY | ORCL_DR        | NOT ALLOWED       |

```
SQL>
```

**:Standby database created successfully:**

~~~~~END~~~~~

---